Seat No.: _____ Enrolment No._____

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER–VIII(NEW) EXAMINATION – WINTER 2022

Subject Code:2180703 **Date:15-12-2022**

Subject Name:Artificial Intelligence

Time:10:30 AM TO 01:00 PM **Total Marks:70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

			Marks
Q.1	**(a)**	Define the term "Artificial Intelligence. What is an AI technique?	**03**
	(b)	What is production system? Discuss the characteristics of a production system.	**04**
	(c)	Explain the State space representation with water jug problem.	**07**
Q.2	**(a)**	Describe heuristic function for the Travelling Salesman Problem.	**03**
	(b)	Differentiate between Breadth First Search and Depth First Search.	**04**
	(c)	Solve the following cryptarithmetic problem using constraint satisfaction. Every letter must be assigned unique digit between 0 to 9.	**07**

```
  B A S E
+ B A L L
  ---------
G A M E S
```

OR

	(c)	Discuss Simulated Annealing search method. Compare it with hill climbing method.	**07**
Q.3	**(a)**	Define the following terms. i. Representational Adequacy ii. Inferential Adequacy iii. Inferential Efficiency	**03**
	(b)	Differentiate between declarative and procedural representation of knowledge.	**04**
	(c)	Explain the algorithm of resolution in predicate logic.	**07**

OR

Q.3	**(a)**	What are the issues in knowledge representation?	**03**
	(b)	Differentiate between Forward Reasoning and Backward Reasoning.	**04**
	(c)	Discuss the different approaches to knowledge representation.	**07**
Q.4	**(a)**	What are the different approaches to reasoning under uncertainties?	**03**
	(b)	List out the property of Monotonic and Non monotonic reasoning.	**04**
	(c)	Explain alpha-beta cut off search with an example. State a case when to do alpha pruning.	**07**

OR

Q.4	**(a)**	Explain the following terms: i. Semantic Nets	**03**

ii. Frames

	(b)	Explain Bayesian Networks.	**04**
	(c)	Explain the minimax algorithm with an example.	**07**
Q.5	**(a)**	What is understanding? What makes understanding hard?	**03**
	(b)	Enlist some applications of Neural Networks.	**04**
	(c)	Write Prolog programs to perform the following with LIST: i) Delete an element from a list ii) Concatenating of two lists	**07**

OR

Q.5	**(a)**	List out the few properties of Prolog programming.	**03**
	(b)	Write a prolog program find maximum number from a list.	**04**
	(c)	Explain steps of Natural Language Processing.	**07**